

```
$ mysql -e "SHOW CREATE TABLE orders" --normalize
```



// 1NF → 2NF → 3NF □□□□□

1NF

□□□

2NF

□□□□□□

3NF

□□□□□□□

□□□□

OLAP □□□

\$ cat schema-plan.md



01 / 
□□□□□□□□

02 / 
1NF    → 2NF → 3NF

03 / 
JOIN □□□

04 / 
OLTP vs OLAP □□□

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PART 01



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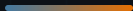
order_id	customer_name	customer_email	product_name	price
1	□□	<u>zhang@example.com</u>	□□□□□	30000
2	□□	<u>zhang@example.com</u>	□□	500
3	□□	<u>li@example.com</u>	□□	1200

□□□□

□□□□□□□□ email □□□□

□□□□

□□□ email□□ UPDATE □□□□



ROOT CAUSE



CHECKPOINT_01

データベースの UPDATE 37 回実行後 3 分経過

A 実行ログを確認

B 実行ログを確認

C 実行ログを確認

PART 02



1NF → 2NF → 3NF → ☐ ☐ ☐ ☐ ☐ ☐

1NF

NO DUPLICATES

BEFORE —

student	courses
	Maths, Maths, Maths, Maths, Maths, Maths, Maths, Maths

AFTER — 1NF

student	course
	Maths
	Maths
	Maths

2NF NOGRAPH

BEFORE — □□□□

order_id	product_id	product_name	qty
1	P1	□□	1
2	P1	□□	3

AFTER — □□□□□

order_id	product_id	qty
1	P1	1
product_id	product_name	
P1	□□	

3NF

NO GRAPH

□□□□□□

student_id → dept_id → dept_name — dept_name — dept_id — — — — —

BEFORE — □□□□

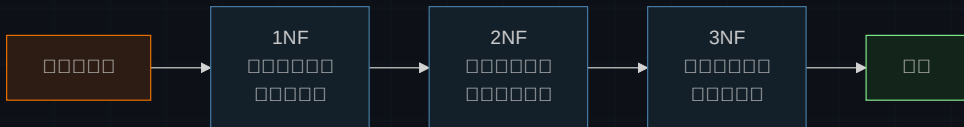
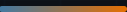
student_id	name	dept_id	dept_name
S1	□□	D1	□□□
S2	□□	D1	□□□

AFTER — □□□□□

student_id	name	dept_id
S1	□□	D1
dept_id	dept_name	
D1	□□□	

□□□□□□□□□□ — □□□□□□□□□□

□ □ □ □ □ □ □



3 columns 3NF 2 columns 6 columns 4 rows

PART 03



□□□□□□JOIN □□□

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JOIN □□
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JOIN □□□□□

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OLTP vs OLAP

OLTP	OLAP	Comparison
Transactional	Analytical	
High write volume	High read volume	
Small number of rows	Large number of rows	
Simple queries	Complex queries	
Real-time data	Historical data	
OLTP	OLAP	

STRATEGY

OLTP is designed for high transaction volume — OLAP is designed for high analytical volume



→



1NF                                   

□ □ □ □ □ □ □ □ □ □ □ □

OLTP □□□□OLAP □□□□□□□□

Schema normalized.

XXXXXXXXXXXX JOIN XXXXX

```
$ mysql -e "EXPLAIN SELECT ..."
```

1NF → 2NF → 3NF → XXXXXXXX